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SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

CHAPTER 1:Introduction and Overview

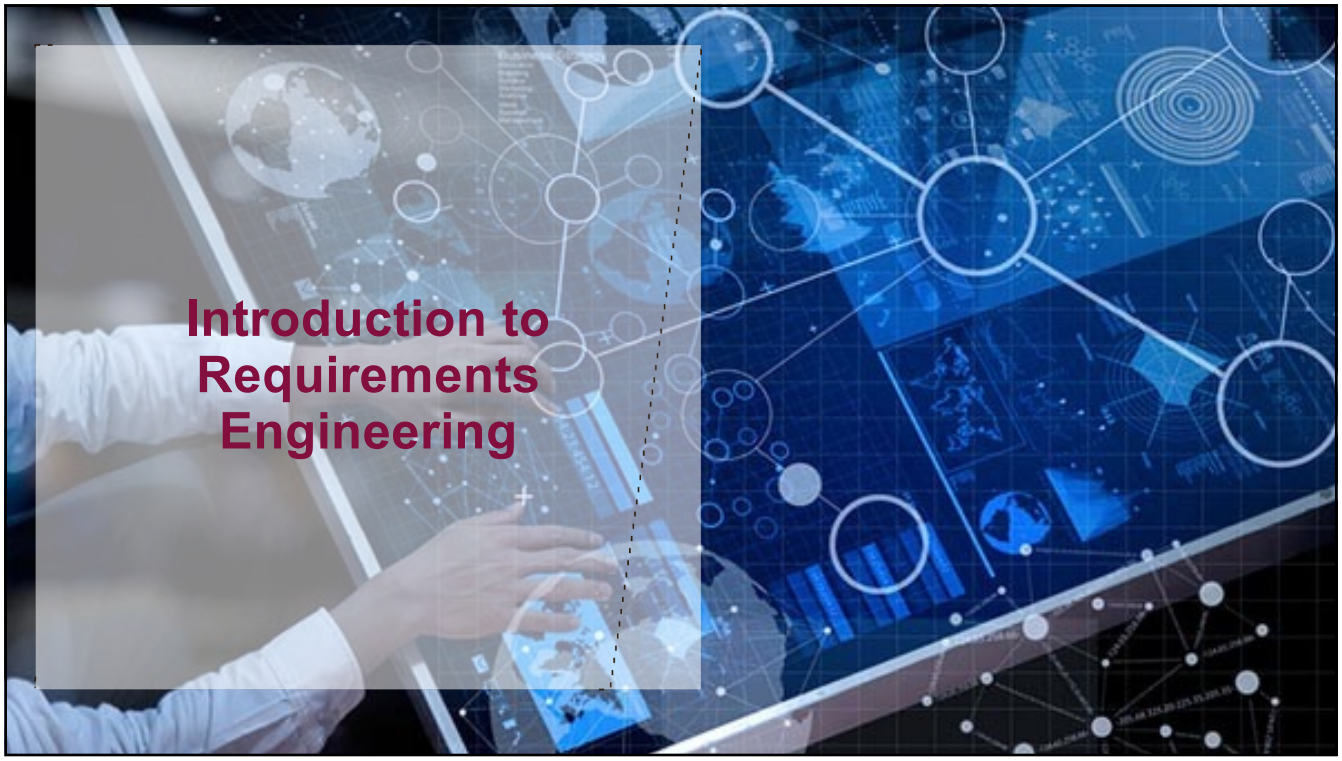
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- Introduction to Requirements Engineering
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Importance of Requirements Engineering

"The hardest single part of building a software system is deciding precisely what to build. No other part of the conceptual works is as establishing the detailed technical requirements, including all interfaces to people, to other software systems. No other part of the work so cripples the resulting system if done wrong. No other part is more difficult to rectify later"

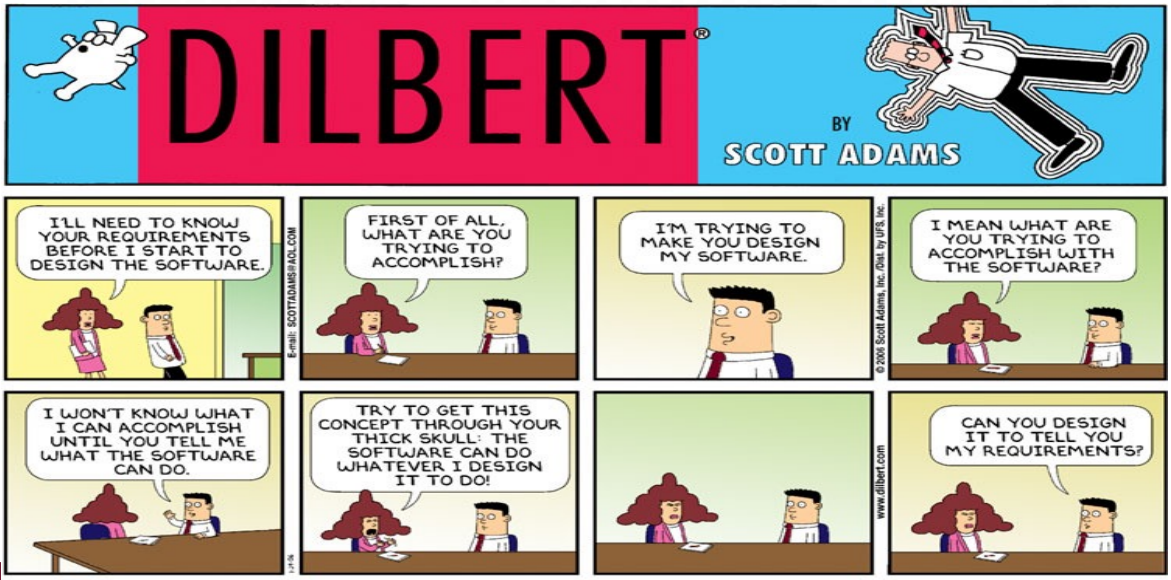
From F.P.Brooks Jr. in "No Silver Bullet – Essence and Accidents of Software Engineering", IEEE Computer, vol.20, no.4, Apr 1987)

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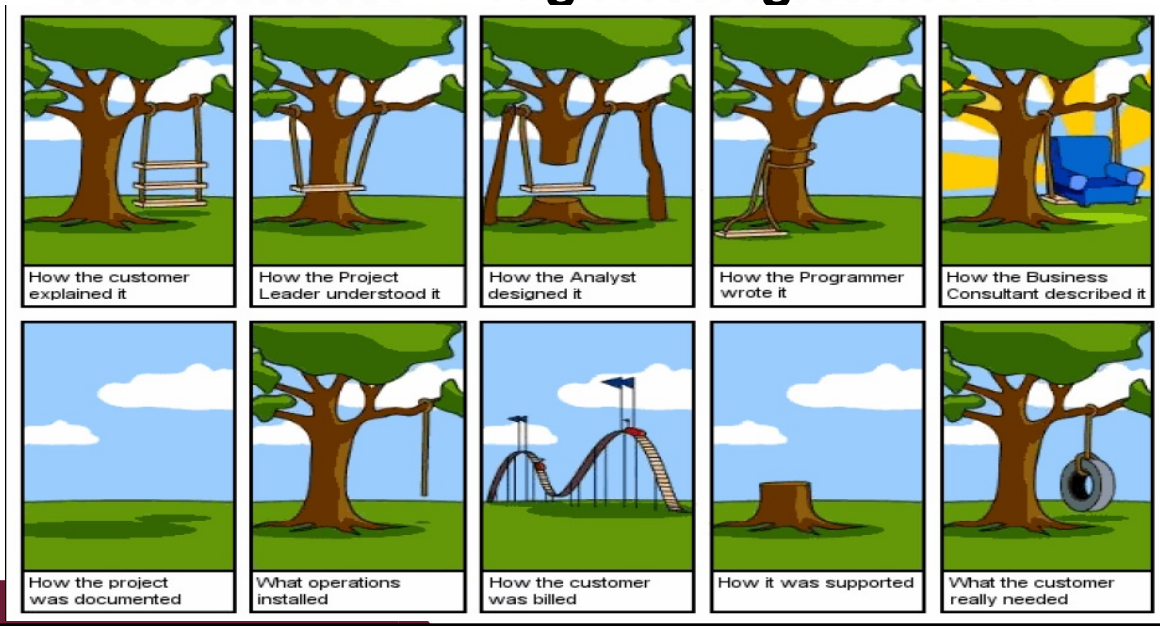
Importance of Requirements Engineering



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Importance of Requirements Engineering



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Requirements

- A documented statement about the proposed system that all stakeholders agree must be made true in order for the customer's problem to be adequately solved
- A capability or property that a system shall have



Stakeholder

- A person or organization who influences a system's requirements or who is impacted by the system.

Note:

- Influence can be categorized as indirect. For example, in some cases stakeholders may have to follow instructions given by their managers and organizations.



Requirements Engineering

The systematic and disciplined approach to the specification and management of requirements with the goal of understanding the stakeholders' desires and needs and minimizing the risk of delivering a system that does not meet these desires and needs.

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Requirements Engineering: What

- Requirements Engineering focuses on defining what needs to be created or improved. These can include:
  a. **Products**: Tangible items delivered to customers.
  b. **Services**: Activities or functions offered to customers.
  c. **Other Deliverables**: Tools, procedures, or devices that help individuals or organizations achieve their objectives.
  d. **Compositions or Components**: Parts or assemblies of products, services, or other deliverables.
- All of these are collectively referred to as **systems**.

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Definition: System

1. General Definition:
A system is a method or principle used to organize and structure elements effectively.

2. Engineering Definition:
In engineering, a system refers to a well-defined and bounded collection of components that work together in a coordinated way to fulfill a specific purpose.

Notes:
1. A system can include other systems or smaller parts known as subsystems.
2. A system can achieve its goals in various ways, such as:
- Installing or using the system at the intended location(s),
- Selling or delivering the system as a product to end-users,
- Offering the system's functions as services through service providers.
3. Systems that combine both software and physical components are called **cyber-physical systems**.
4.Systems that integrate software, hardware, human roles, and organizational structures are known as **socio-technical systems**.

Requirements Engineering: Why

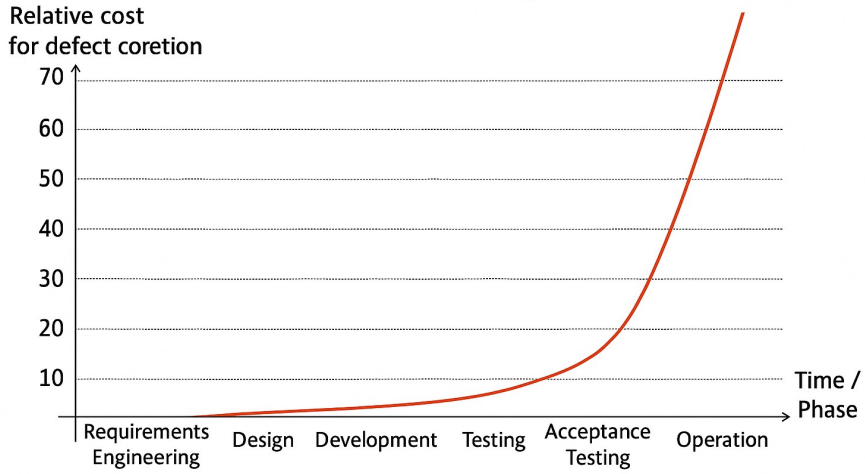
Effective Requirements Engineering (RE) enhances the development and evolution of a system by:

1.Minimizing the risk of building an incorrect system
- RE helps prevent failures or expensive changes during later development phases
- Refer to the cost implications of fixing defects

2. Improving clarity around the problem to be solved
- RE reduces the cognitive load in grasping the problem the system aims to address and aids in evaluating potential solutions

1.Providing a foundation for estimating development time and costs
2.Serving as a necessary step for conducting proper system testing

Cost of correcting defects



Source: Barry Boehm, *Software Engineering* Economics, Prentice Hall, 1981

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Symptoms and Reasons for inadequate RE

- Common signs of ineffective Requirements Engineering (RE) include: Requirements that are missing, vague, or inaccurate

- Key contributing factors are:

1. Starting system development hastily due to tight project timelines or schedule pressure.
2. Poor communication among involved parties, especially between stakeholders and developers or within stakeholder groups.
3. Assuming that requirements are obvious or self-explanatory, which often leads to misunderstandings.
4. Lack of proper training or insufficient skills in RE among team members responsible for conducting it.

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Requirements Engineering: Where

- RE can be applied to requirements for any kind of system.
- The dominant application case for RE today is represented by systems in which software plays a major role.
- Such systems typically consist of
  - Software components
  - Physical elements (technical products, computing hardware, devices, sensors, etc)
  - Organizational elements (persons, positions, business processes, legal and compliance issues, etc)

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Requirements Engineering: Where

- Systems that integrate both software and physical components are referred to as cyber-physical systems.
- Systems that encompass software, hardware, human, and organizational elements are known as socio-technical systems.

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Four Key Task in RE



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The Role and Tasks of a Requirements Engineer

Nature of the Role

- A Requirements Engineer is not strictly a job title but rather a role that individuals may perform.
- Those in this role typically:
  - Elicit, document, validate, and/or manage requirements as part of their responsibilities.
  - Possess in-depth knowledge of Requirements Engineering (RE), enabling them to define RE processes, choose suitable practices, and apply them effectively.
  - Bridge the gap between identifying a problem and implementing potential solutions.

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The Role and Tasks of a Requirements Engineer

Scope and Applicability

- The role of a Requirements Engineer is embedded within various job functions defined by organizations.
- In practice, individuals such as business analysts, application specialists, product owners, systems engineers, and developers may perform the duties of a Requirements Engineer.
- RE knowledge and skills are also valuable to other professionals, including designers, testers, system architects, and Chief Technology Officers (CTOs).

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Foundational skill set that a Requirements Engineer must learn

A Requirements Engineer must develop the following essential competencies:

i. Master the fundamental principles of Requirements Engineering (RE) that apply to all RE tasks, activities, and practices.
ii. Document requirements in a variety of forms suitable for the specific context and situation.
iii. Elaborate requirements, including elicitation and validation using the most appropriate practices for the given situation.
iv. Define and apply suitable RE processes effectively.
v. Manage and maintain existing requirements efficiently.
vi. Utilise appropriate tool support to assist in performing RE tasks.

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Foundational skill set that a Requirements Engineer must learn

In addition to technical and analytical abilities, a Requirements Engineer must possess key interpersonal skills, often referred to as soft skills:

- The ability to listen actively, moderate discussions, negotiate effectively, and mediate conflicts.
- Empathy towards stakeholders and openness to understanding and considering the needs and ideas of others.

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Types of Requirements

Functional Requirements

- Concern a result or behavior that shall be provided by a function of a system
- This includes requirements for data or the interaction of a system with its environment

Constraint

- Limitations or restrictions on the system's design, development, or implementation.
- These can be imposed by various factors, including technology, budget, or time.
- Example: The project must be completed within 6 months

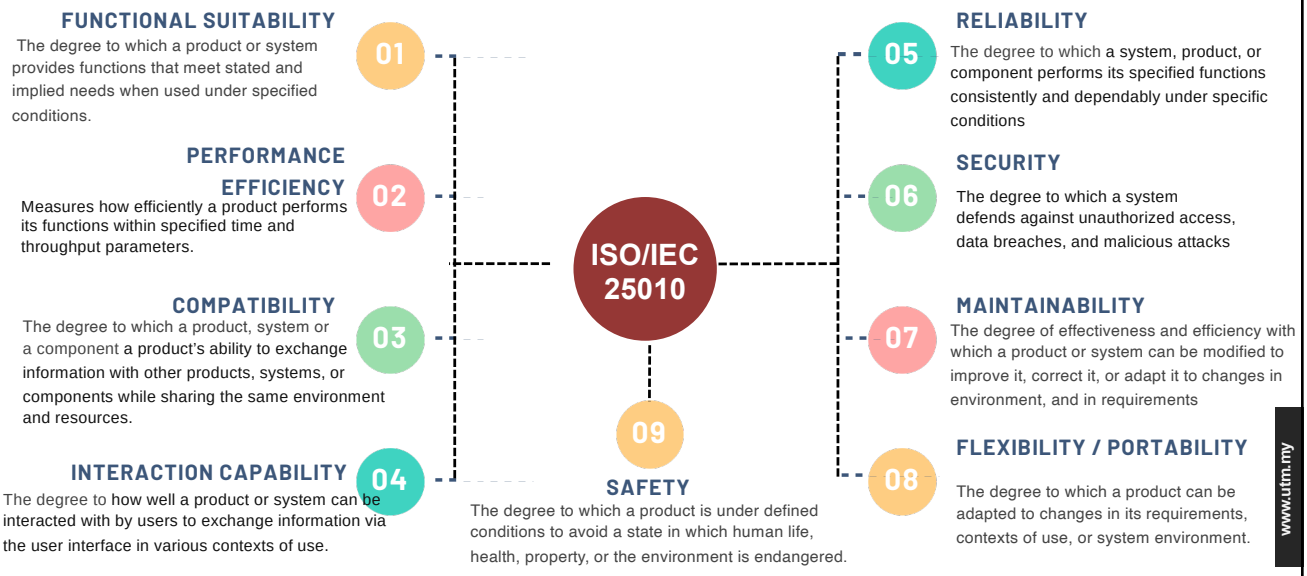
Quality Requirements

- Pertain to quality concerns that are not covered by functional requirements
- Highly influence the system architecture
- Such as performance, availability, security, or reliability

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Quality Requirements



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Sub-Quality Requirements

1 - FUNCTIONAL SUITABILITY

1.FUNCTIONAL COMPLETENESS
2.FUNCTIONAL CORRECTNESS
3.FUNCTIONAL APPROPRIATENESS

2 - PERFORMANCE EFFICIENCY

1.TIME BEHAVIOUR
2.RESOURCE UTILIZATION
3.CAPACITY

3 - COMPATIBILITY

1.CO-EXISTENCE
2.INTEROPERABILITY

4 - INTERACTION CAPABILITY

1.APPROPRIATENESS RECOGNIZABILITY
2.LEARNABILITY
3.OPERABILITY
4.USER ERROR PROTECTION
5.USER ENGAGEMENT
6.INCLUSIVITY
7.USER ASSISTANCE
8.SELF-DESCRIPTIVENESS

5 - RELIABILITY

1.FAULTLESSNESS
2.AVAILABILITY
3.FAULT TOLERANCE
4.RECOVERABILITY

6 - SECURITY

1.CONFIDENTIALITY
2.INTEGRITY
3.NON-REPUDIATION
4.ACCOUNTABILITY
5.AUTHENTICITY
6.RESISTANCE

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Sub-Quality Requirements (cont.)

7 - MAINTAINABILITY

1.MODULARITY
2.REUSABILITY
3.ANALYSABILITY
4.MODIFIABILITY
5.TESTABILITY

8 - FLEXIBILITY

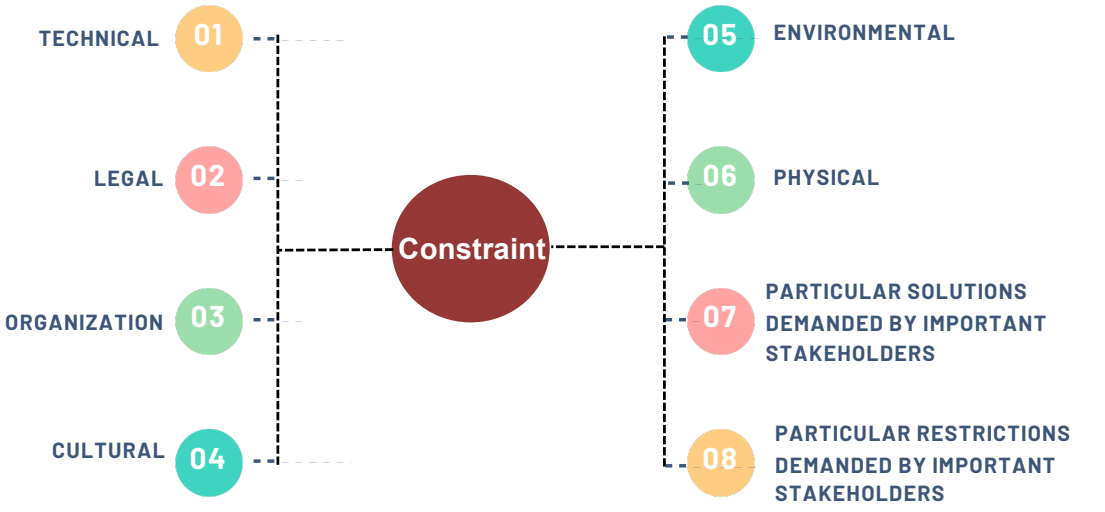
1.ADAPTABILITY
2.SCALABILITY
3.INSTALLABILITY
4.REPLACEABILITY

9 - SAFETY

1.OPERATIONAL CONSTRAINT
2.RISK IDENTIFICATION
3.FAIL SAFE
4.HAZARD WARNING
5.SAFE INTEGRATION

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Constraints Category



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Different Sort of Requirements

Business Requirements
Focus on the business aims, objectives, and needs that the organization intends to achieve by using a system (or a set of systems).

Domain Requirements
- Specify the essential properties of the domain.
- Define the necessary characteristics of a socio-technical or cyber-physical system within its specific domain.

Stakeholder Requirements
Capture the stakeholders' needs and expectations, outlining what should be fulfilled by building the system, as viewed from the stakeholders' standpoint.

Requirements

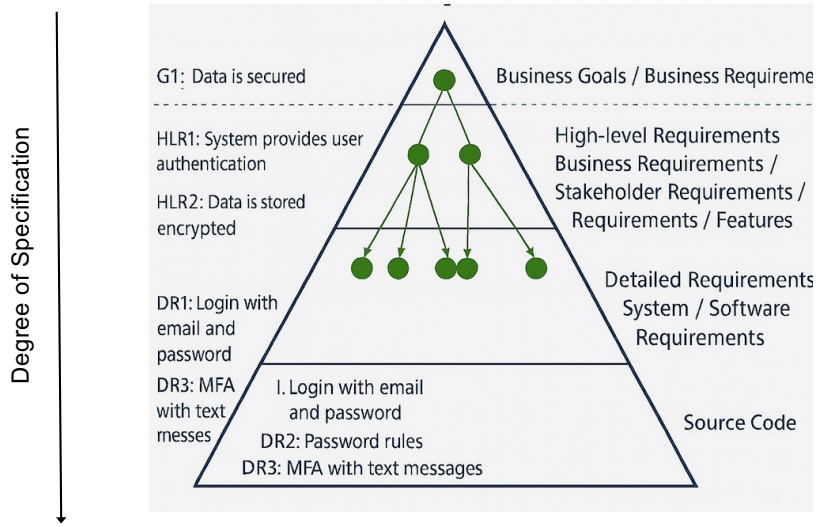
User Requirements
- Represent a subset of stakeholder requirements.
- Address the needs and desires of the system's users.

System Requirements
- Explain how the system should operate and behave at the interface between the system and its environment, ensuring it meets stakeholders' needs and expectations.
- For systems that are purely software-based, these are referred to as software requirements.

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Levels of Specification



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